

Preliminary PhD Research Evidence

Semantic Utility-Aware Compression for Wildfire-Centric Earth Observation

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Research problem

Under restricted satellite downlink, which image information should be sent first? This project asks whether wildfire-aware ranking of learned image tokens can retain mission-relevant evidence more efficiently than distortion-optimised compression. Wildfire remains the primary mission. The central hypothesis is **not yet confirmed**: JPEG2000 leads in the current matched-ceiling study.

System and pre-scoring design

Post-fire Sentinel-2 RGB is analysed for wildfire utility, encoded as VQ-VAE tokens, ranked, serialised with a retention mask, transmitted and reconstructed. The one-time official-validation comparison covers **60 event pairs / 120 images**, with one burn and one nearby negative chip per event. Adaptation used a separate 600-event source-training selection (480 gradient-training, 120 internal validation). All methods were constrained to **at most 1,200 actual serialised bytes per native 224 x 224 image**; JPEG/JPEG2000 settings were selected for original-image PSNR under that ceiling. Payloads are under a common cap, not identical in size.

What was measured

SUS is a model-dependent 0-100 combination of detector confidence, object, relevance and important-region retention. Proxy Dice compares predicted regions with a separate dNBR-derived mask; negative predicted-positive area indicates false-alarm burden on nominally negative chips, but is not a calibrated false-positive rate. PSNR, SSIM and actual serialised bytes measure visual and communication performance. No single endpoint establishes operational wildfire benefit.

Main result on 60 burn chips

Method	SUS /100	Proxy Dice	PSNR (dB)	SSIM	Mean bytes
JPEG	92.72	0.519	27.97	0.941	1,116.7
JPEG2000 RDO	95.12	0.533	27.18	0.913	1,177.5
Base VQ-VAE	70.85	0.286	19.87	0.644	1,140.8
Adapted VQ-VAE	83.28	0.469	20.78	0.636	1,155.4

Paired differences (adapted VQ-VAE minus comparator)

Versus base: **+12.43 SUS** (95% event-bootstrap CI +8.74 to +16.31) and **+0.183 proxy Dice** (+0.122 to +0.245).

Versus JPEG2000: **-11.85 SUS** (CI -15.14 to -8.72) and **-0.064 proxy Dice** (-0.113 to -0.018). Ten thousand event-paired bootstrap resamples were used.

Interpretation and research programme

What the result does and does not establish

Training adaptation changes wildfire-relevant retention and improves the VQ-VAE relative to its base checkpoint. It does **not** show an advantage over JPEG2000. Negative-chip predicted-positive area rose from 0.225 to 0.509; the paired increase is +0.284 (CI +0.202 to +0.365). The separately frozen 18-event development gate failed, and the 37-pair screened test cohort remains sealed and unscored.

Validity limits

Validation event IDs are distinct from selected training and development IDs, but every selected validation country occurs in training; patch-level IDs do not prove scene-footprint separation. The burn mask is a quantised dNBR spectral proxy, not a manually checked burn-scar boundary. Five images required reprojection with uncovered edges excluded. SUS and detector retention share a detector, so they are not independent endpoints. No LPIPS or measured onboard energy result is claimed here.

Three focused PhD questions

- 1. Measurement:** Can utility estimates be validated against independent geospatial fire annotations, with calibrated uncertainty and negative-scene specificity?
- 2. Representation:** At matched serialised byte budgets, can learned token prioritisation improve both SUS and independent-mask Dice over JPEG2000 without increasing false alarms?
- 3. Mission feasibility:** When do any measured utility gains justify onboard latency, memory, compute and downlink cost?

First-year validation sequence

Secure a licensed independently annotated Sentinel-2 cohort; verify exact event, footprint and geographic separation from known training; fix preprocessing, byte accounting, endpoints and thresholds before scoring; run a paired codec comparison across byte budgets with both positive and negative scenes. Report negative findings if the conventional baselines remain stronger. Keep the old failed gate and sealed cohort unchanged.

Bounded four-year trajectory

Year 1: provenance, independent labels and frozen evaluation protocol. **Year 2:** representation and token-ranking studies confined to training/development data, with false-alarm checks. **Year 3:** measured edge-device inference and serialisation costs alongside clearly labelled link simulations. **Year 4:** one untouched external replication after predeclared advancement, papers including negative findings, and thesis synthesis. If JPEG2000 remains stronger, identifying the conditions and measurement limits behind that failure is still a valid research outcome.

Reproducibility note

Frozen selection, hashes, per-image rows and paired statistics: [results/future_satellite_cohort_audit/ecofirebias_official_validation/](#). Source dataset: EcoFireBias / wildfire_global, Hugging Face revision 39f331e50458fba3669837d69fc2fdddbb1d1d69. This annex supersedes older positive-sounding evidence summaries; the existing proposal was not edited.